

Spotify Music Analysis

Practice Questions & Solutions

Q1. Retrieve the names of all tracks that have more than 1 billion streams.

Solution:

```
SELECT
    artist,
    track,
    channel,
    stream
FROM spotify
WHERE stream > 10000000000
ORDER BY stream DESC;
```

Q2. List all albums along with their respective artists.

Solution:

```
WITH agg AS
(
    SELECT
        DISTINCT
            album,
            artist
    FROM spotify
    ORDER BY 1
)
SELECT
    album,
    STRING_AGG(artist, ', ' ) AS "Artist Name"
FROM agg
GROUP BY 1
ORDER BY 1;
```

Q3. Get the total number of comments for tracks where licensed = TRUE.

Solution:

```
SELECT
    SUM(comments) AS "Total Comment"
FROM spotify
WHERE licensed IN (TRUE);
```

Q4. Find all tracks that belong to the album type single.

Solution:

```
SELECT
    *
FROM spotify
WHERE album_type IN ('single');
```

Q5. Count the total number of tracks by each artist.

Solution:

```
WITH track AS
(
    SELECT
        DISTINCT
            artist,
            track
    FROM spotify
    ORDER BY 1
)
SELECT
    artist,
    COUNT(track) AS "Total Track"
FROM track
GROUP BY 1
ORDER BY 2 DESC;
```

Q6. Calculate the average danceability of tracks in each album.

Solution:

```
WITH dance AS
(
SELECT
    DISTINCT
        album,
        track,
        danceability
FROM spotify
ORDER BY 1
)
SELECT
    album,
    track,
    ROUND(AVG(danceability),2) AS AVG
FROM dance
GROUP BY 1,2
ORDER BY 1,2 DESC;
```

Q7. Find the top 5 tracks with the highest energy values.

Solution:

```
SELECT
    track,
    MAX(energy)
FROM spotify
GROUP BY 1
ORDER BY 2 DESC
LIMIT 5;
```

Q8. List all tracks along with their views and likes where official_video = TRUE.

Solution:

```
SELECT
    track,
    SUM(views) AS "Total Views",
    SUM(likes) AS "Total Likes"
FROM spotify
WHERE official_video IN (TRUE)
GROUP BY 1
ORDER BY 2 DESC;
```

Q9. For each album, calculate the total views of all associated tracks.

Solution:

```
WITH view AS
(
    SELECT
        DISTINCT
            artist,
            track,
            views
    FROM spotify
)
SELECT
    track,
    STRING_AGG(artist, ', '),
    MAX(views) AS views
FROM view
GROUP BY 1
ORDER BY 3 DESC;
```

Q10. Retrieve the track names that have been streamed on Spotify more than YouTube.

Solution:

```
SELECT
    *
FROM spotify
WHERE most_played ON IN ('Spotify');
```

Q11. Find the top 3 most-viewed tracks for each artist using window functions.

Solution:

```
WITH top3 AS
(
SELECT
    artist,
    track,
    views,
    ROW_NUMBER() OVER(PARTITION BY artist ORDER BY views DESC) AS rno
FROM spotify
)
SELECT
    artist,
    track,
    views
FROM top3
WHERE rno <= 3;
```

Q12. Write a query to find tracks where the liveness score is above the average.

Solution:

```
SELECT
    artist,
    track,
    liveness
FROM spotify
WHERE liveness > (SELECT AVG(liveness) FROM spotify)
ORDER BY 3 DESC;
```

Q13. Use a WITH clause to calculate the difference between the highest and lowest energy values for tracks in each album.

Solution:

```
WITH diff AS
(
SELECT
    album,
```

```

    MAX(energy) AS max_energy,
    MIN(energy) AS min_energy
FROM spotify
GROUP BY 1
ORDER BY 1
)
SELECT
    *,
    ROUND(max_energy - min_energy,2) AS energy_diff
FROM diff
ORDER BY 1;

```

Q14. Find tracks where the energy-to-liveness ratio is greater than 1.2.

Solution:

```

SELECT
    track,
    artist,
    ROUND(energy/liveness,2) AS energyliveness
FROM spotify
WHERE energyliveness > 1.2;

```

Q15. Calculate the cumulative sum of likes for tracks ordered by the number of views, using window functions.

Solution:

```

WITH cum_likes AS
(
SELECT
    track,
    MAX(views) AS views,
    MAX(likes) AS likes
FROM spotify
GROUP BY 1
ORDER BY 2 DESC
)
SELECT
    *,

```

```
SUM(likes) OVER(ORDER BY views DESC) AS cum_likes  
FROM cum_likes;
```